

Homework 1

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Problem 1

(a) $2u = [4, -2, 6]^T$, so $2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$

Dot that with w : $(3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$

$(2u - v) \cdot w = 24$

(b) $(1/2)w = [-2, 1, 3]^T$, so $v + (1/2)w = [1 - 2, 4 + 1, -2 + 3]^T = [-1, 5, 1]^T$

$||v + (1/2)w|| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3}$

$3\sqrt{3} \approx 5.20$

(c) I need the dot product and both lengths, since $\cos \theta = (u \cdot w) / (||u|| ||w||)$.

$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 12$

$||u|| = \sqrt{4 + 1 + 9} = \sqrt{14}$, $||w|| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$

$||u|| ||w|| = \sqrt{14} \cdot 2\sqrt{14} = 2(14) = 28$

$\cos \theta = 12/28 = 3/7$, so $\theta = \arccos(3/7)$

$\theta \approx 64.62^\circ$

Problem 2

(a) AB is undefined. A is 3×3 so it has 3 columns, B is 2×3 so it has 2 rows, $3 \neq 2$.

BA is fine: $(2 \times 3)(3 \times 3) = 2 \times 3$.

row 1: $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$

row 2: $[3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 2, -1]$

$BA = [2 \ -9 \ -2]$

$[7 \ 2 \ -1]$

(b) A is 3×3 , v is 3×1 , so Av is 3×1 .

$Av = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$

vA is undefined: v has 1 column, A has 3 rows, $1 \neq 3$.

(c) Aw is undefined: A has 3 columns, w is 1×3 so it has 1 row, $3 \neq 1$.

wA is $(1 \times 3)(3 \times 3) = 1 \times 3$.

$wA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4 \ -6 \ 1]$

(d) vw is $(3 \times 1)(1 \times 3) = 3 \times 3$.

$vw = [1(2) \ 1(0) \ 1(-1)] [2 \ 0 \ -1]$

$[-2(2) \ -2(0) \ -2(-1)] = [-4 \ 0 \ 2]$

$[3(2) \ 3(0) \ 3(-1)] [6 \ 0 \ -3]$

wv is $(1 \times 3)(3 \times 1) = 1 \times 1$, just the dot product: $wv = 2(1) + 0(-2) + (-1)(3) = -1$

(e) $A^2 = AA$, both 3×3 .

row 1: $[2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$

$$\text{row 2: } [1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$$

$$\text{row 3: } [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 12, -7]$$

$$A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 12 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 12 & -7 \end{bmatrix}$$

(f) $B^2 = BB$ is undefined. B is 2×3 , not square: the first B has 3 columns, the second has 2 rows, $3 \neq 2$.

Problem 3

(a) Augmented matrix and row reduce:

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1 \quad \begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & 0 \end{bmatrix}$$

$$R_2 \rightarrow -R_2 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2 \quad \begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 12 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 6/5 \end{bmatrix}$$

Three pivots, so one solution. Back substituting, $x_3 = 6/5$, then

$$x_2 = -4 + 3(6/5) = -4 + 18/5 = -2/5$$

$$x_1 = -1 - 2(-2/5) + 6/5 = -1 + 4/5 + 6/5 = 1$$

$$x = [1, -2/5, 6/5]^T$$

$$(b) \quad [1 \ 2 \ -1 \ | \ 4]$$

$$[2 \ 5 \ 1 \ | \ 6]$$

$$[3 \ 7 \ 0 \ | \ 10]$$

$$R_2 \rightarrow R_2 - 2R_1 \quad [1 \ 2 \ -1 \ | \ 4]$$

$$R_3 \rightarrow R_3 - 3R_1 \quad [0 \ 1 \ 3 \ | \ -2]$$

$$[0 \ 1 \ 3 \ | \ -2]$$

$$R_3 \rightarrow R_3 - R_2 \quad [1 \ 0 \ -7 \ | \ 8]$$

$$R_1 \rightarrow R_1 - 2R_2 \quad [0 \ 1 \ 3 \ | \ -2]$$

$$[0 \ 0 \ 0 \ | \ 0]$$

Column 3 has no pivot, so z is free. Let $z = t$.

Row 2: $y = -2 - 3t$. Row 1: $x = 8 + 7t$.

Infinitely many solutions, one for each value of t .

$$[x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}$$

Problem 4

(a) Let x_1, x_2, x_3 be the masses, in kg, of Alloy 1, Alloy 2, Alloy 3 that get used. The 100 kg product holds 36 kg copper, 44 kg zinc, 20 kg nickel, and nothing is lost,

so for each metal the mass going in equals the mass coming out.

copper: $0.5x_1 + 0.2x_2 + 0.4x_3 = 36$

zinc: $0.3x_1 + 0.7x_2 + 0.2x_3 = 44$

nickel: $0.2x_1 + 0.1x_2 + 0.4x_3 = 20$

(b) Multiply each row by 10 to clear the decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c) $[5 \ 2 \ 4 \mid 360]$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_1 \rightarrow R_1 - 2R_3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1 \quad [1 \ 0 \ -4 \mid -40]$$

$$R_3 \rightarrow R_3 - 2R_1 \quad [0 \ 7 \ 14 \mid 520]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2 \quad [1 \ 0 \ -4 \mid -40]$$

$$R_3 \rightarrow R_3 - R_2 \quad [0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 0 \ 10 \mid 1440/7]$$

$$R_3 \rightarrow (1/10)R_3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 0 \ 1 \mid 144/7]$$

$$x_3 = 144/7 \approx 20.57, x_2 = 520/7 - 2(144/7) = 232/7 \approx 33.14$$

$$x_1 = -40 + 4(144/7) = -40 + 576/7 = 296/7 \approx 42.29$$

use about 42.29 kg of Alloy 1, 33.14 kg of Alloy 2, and 20.57 kg of Alloy 3.

Problem 5

$$(a) \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2 \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

(b) $\text{rank}(A) = 2$, because the RREF has pivots in columns 1 and 3, so two pivots (two nonzero rows).

(c) The same operations on $[A \mid b]$ leave the bottom row as $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$.
 0 solutions: possible. If $b_3 \neq b_1 + b_2$, for example $b = [0, 0, 1]^T$, that row says $0 = 1$, so the system is inconsistent.

Exactly 1 solution: impossible. Column 2 never holds a pivot, so x_2 is free whenever

the system is consistent. $\text{rank}(A) = 2 < 3$ unknowns, so the solution is never unique.

Infinitely many: possible. If $b_3 = b_1 + b_2$, say $b = [1, 3, 4]^T$ or $b = 0$, the system

is consistent and the free variable x_2 gives infinitely many solutions.

Problem 6

(a) Vectors as columns, then row reduce:

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1 \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2 \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Two pivots for three vectors, so there is a free variable and the set is dependent.

The relation is $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$, and v_1, v_2 are not multiples of each other so they still span two dimensions.

Dependent, and the span is a plane through the origin in $\mathbb{R}^3$.

(b) $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \quad \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$$

Three pivots for three vectors, so $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ only when $c_1 = c_2 = c_3 = 0$, which is what independent means, and 3 of them fill $\mathbb{R}^3$. Independent, and the span is all of $\mathbb{R}^3$.
